

MCQs FROM CHAPTER: SORTING & TIME COMPLEXITY

(Bubble Sort, Selection Sort, Insertion Sort, Complexity)

◆ GENERAL INTRODUCTION & CONCEPT OF SORTING

1. Sorting is the process of
 - A) Searching elements
 - B) Deleting elements
 - C) Ordering or arranging elements
 - D) Encrypting data
2. Sorting numbers in increasing order is known as
 - A) Descending order
 - B) Ascending order
 - C) Random order
 - D) Linear order
3. Sorting strings alphabetically from a-z is an example of
 - A) Numerical sorting
 - B) Length-based sorting
 - C) Lexicographical sorting
 - D) Reverse sorting
4. Seats in an examination hall are generally arranged based on
 - A) Height
 - B) Weight
 - C) Roll number
 - D) Age
5. Sorting a dictionary helps in
 - A) Printing words
 - B) Searching faster
 - C) Increasing memory
 - D) Reducing pages
6. Searching in an unsorted dictionary would be
 - A) Fast
 - B) Efficient
 - C) Tedious
 - D) Automatic
7. Extra time taken during sorting is called
 - A) Delay
 - B) Latency
 - C) Overhead
 - D) Processing loss
8. Sorting is an important area of study in
 - A) Biology

- B) Mathematics
 - C) Computer Science
 - D) Physics
9. This chapter discusses how many sorting methods?
- A) One
 - B) Two
 - C) Three
 - D) Four
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◆ **BUBBLE SORT**

10. Bubble sort works by repeatedly
- A) Selecting minimum element
 - B) Inserting elements
 - C) Comparing adjacent elements
 - D) Dividing the list
11. In bubble sort, swapping means
- A) Deleting elements
 - B) Changing positions of elements
 - C) Copying elements
 - D) Reversing the list
12. One full traversal of the list in bubble sort is called
- A) Step
 - B) Cycle
 - C) Pass
 - D) Phase
13. For a list of n elements, bubble sort makes
- A) n passes
 - B) $n+1$ passes
 - C) $n-1$ passes
 - D) n^2 passes
14. After each pass of bubble sort in ascending order
- A) Smallest element reaches start
 - B) Largest element reaches correct position
 - C) Middle element is fixed
 - D) List is fully sorted
15. The name “Bubble sort” comes from
- A) Use of bubbles
 - B) Largest element bubbling up
 - C) Smallest element sinking
 - D) Random swapping
16. In bubble sort, already sorted elements are
- A) Rechecked every time

- B) Removed from list
 - C) Ignored in further passes
 - D) Copied to another list
17. Bubble sort compares elements
- A) Randomly
 - B) Pairwise adjacent
 - C) Only first and last
 - D) Alternate positions
18. If no swapping occurs in a pass, it means
- A) List is empty
 - B) List is partially sorted
 - C) List is already sorted
 - D) Error in algorithm
19. Bubble sort can be optimized by
- A) Adding recursion
 - B) Stopping when no swap occurs
 - C) Increasing passes
 - D) Sorting twice
20. In bubble sort, comparison index j goes from
- A) 0 to n
 - B) 0 to $n-1$
 - C) 0 to $n-i-1$
 - D) i to n
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◆ BUBBLE SORT ALGORITHM & PROGRAM

21. Bubble sort uses how many loops?
- A) One
 - B) Two
 - C) Three
 - D) None
22. In bubble sort algorithm, variable i represents
- A) Element value
 - B) Pass count
 - C) Swap counter
 - D) Index of largest element
23. In Python bubble sort, the outer loop controls
- A) Comparisons
 - B) Swaps
 - C) Number of passes
 - D) Sorted output
24. In the Python code, `range(0, n-i-1)` is used because
- A) List size changes

- B) Last i elements are already sorted
 - C) To avoid errors
 - D) For faster execution
25. Bubble sort can be used to sort in descending order by
- A) Removing swaps
 - B) Reversing output
 - C) Changing comparison operator
 - D) Changing loop direction
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◆ SELECTION SORT

26. Selection sort divides the list into
- A) Three parts
 - B) Two parts: sorted and unsorted
 - C) Equal halves
 - D) Random segments
27. Initially, in selection sort, the sorted list is
- A) Full
 - B) Half
 - C) Empty
 - D) Random
28. Selection sort makes how many passes for n elements?
- A) n
 - B) $n-1$
 - C) $n+1$
 - D) n^2
29. In selection sort, during each pass
- A) Largest element is selected
 - B) Random element is selected
 - C) Smallest element is selected
 - D) Middle element is selected
30. The smallest element found is swapped with
- A) Last element
 - B) Rightmost element
 - C) Leftmost unsorted element
 - D) Middle element
31. After each pass of selection sort
- A) One element is removed
 - B) One element is added to sorted list
 - C) Entire list is sorted
 - D) List size doubles
32. Selection sort continues until
- A) All elements are compared

- B) $n-1$ elements are placed
 - C) n elements are swapped
 - D) List becomes empty
33. The last element in selection sort
- A) Needs sorting
 - B) Is already in place
 - C) Is swapped twice
 - D) Is ignored
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◆ SELECTION SORT ALGORITHM & PROGRAM

34. Variable min in selection sort stores
- A) Maximum value
 - B) Index of smallest element
 - C) Sorted count
 - D) Pass number
35. The flag variable in selection sort is used to
- A) Count comparisons
 - B) Detect swapping condition
 - C) Print output
 - D) Stop execution
36. Selection sort compares elements using
- A) One loop
 - B) Nested loops
 - C) Recursion
 - D) Conditional branching only
37. In selection sort Python program, swapping happens
- A) Always
 - B) Never
 - C) Only if flag is set
 - D) Randomly
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◆ INSERTION SORT

38. In insertion sort, the list is divided into
- A) Sorted and reversed
 - B) Odd and even
 - C) Sorted and unsorted
 - D) Random partitions
39. In insertion sort, elements are inserted
- A) At the end
 - B) At the beginning

- C) At appropriate position
 - D) Randomly
40. In insertion sort, sorted list is traversed
- A) Forward
 - B) Backward
 - C) Randomly
 - D) Alternately
41. In the first pass of insertion sort
- A) Sorted list is empty
 - B) Sorted list has one element
 - C) Sorted list has two elements
 - D) Unsorted list is empty
42. In insertion sort, elements are shifted
- A) Left
 - B) Right
 - C) Both directions
 - D) Not shifted
43. Insertion sort is similar to
- A) Arranging books
 - B) Playing cards arrangement
 - C) Filling forms
 - D) Queue management
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◆ INSERTION SORT ALGORITHM & PROGRAM

44. In insertion sort algorithm, variable temp stores
- A) Index
 - B) Sorted value
 - C) Current element to insert
 - D) Loop counter
45. In insertion sort, comparison continues while
- A) $j < n$
 - B) $j \geq 0$ and $\text{numList}[j] > \text{temp}$
 - C) $i < n$
 - D) temp is smallest
46. After shifting elements, insertion is done using
- A) $\text{numList}[j]$
 - B) $\text{numList}[i]$
 - C) $\text{numList}[j+1]$
 - D) $\text{numList}[\text{temp}]$
47. Insertion sort uses how many loops?
- A) One
 - B) Two

- C) Three
 - D) None
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◆ **TIME COMPLEXITY OF ALGORITHMS**

48. Time complexity refers to
- A) Memory used
 - B) Code length
 - C) Processing time taken
 - D) Output size
49. Time complexity becomes significant when
- A) Data is small
 - B) Data is huge
 - C) Program is short
 - D) Output is small
50. An algorithm without loops has time complexity
- A) n
 - B) n^2
 - C) 1
 - D) $\log n$
51. Such algorithms are called
- A) Linear time
 - B) Quadratic time
 - C) Constant time
 - D) Exponential time
52. An algorithm with a single loop from 1 to n has
- A) Time complexity n
 - B) Time complexity n^2
 - C) Constant time
 - D) Zero time
53. Nested loops result in time complexity
- A) n
 - B) $2n$
 - C) n^2
 - D) $n \log n$
54. If an algorithm has both single and nested loops
- A) Single loop dominates
 - B) Nested loop dominates
 - C) Both ignored
 - D) Average taken
55. Bubble sort time complexity is
- A) n
 - B) n^2

- C) 1
- D) $\log n$

56. Selection sort time complexity is

- A) n
- B) n^2
- C) $n \log n$
- D) 1

57. Insertion sort time complexity is

- A) n
- B) n^2
- C) $\log n$
- D) Constant